14 Solve the simultaneous equations

$$4x + 6y = 5$$
$$7x + 5y = -10.5$$

(1) 4x + 6y = 5; (2) 7x + 5y = -10.5

(1) × 5: 20x + 30y = 25

(2) × 6: 42x + 30y = -63

1 mark
Eliminate y

Subtract: 22x = -88, so x = -4.

1 mark
x=−4

Substitute: 4(-4) + 6y = 5

1 mark
Substitute x

-16 + 6y = 5, so6y = 21 and y = 3.5.

1 mark
y=3.5

$x =$ .....

$y =$ .....

(Total for Question 14 is 4 marks)